No More 'Flying Blind' with Catheters

2026 BME Clinical Immersion Program

Antranig Baghdassarian

Need & Inspiration

Two big challenges in interventional radiology:

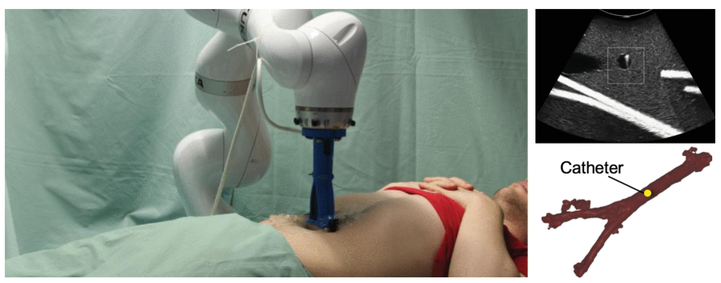
- Catheter navigation
- Guidance through the intricate geometry of the patient's vascular system.

Imaging Success Criteria

- ≤ radiation than C-arm X-ray
- ≤ 1 min per scan
- Resolution ≥ current methods
- Live position tracking

Current Imaging Solutions

1. **X-ray (C-arm)**
   - ✓ Widely used
   - ✓ Fast imaging time
   - X Clinically significant radiation[1,2]
   - X Contrast can cause acute kidney injury[6]
   - X 2D image of 3D object, blocking some vessels[3,5]
2. **3D Fusion Guidance**
   - ✓ Reconstitute 3D image of body from CBCT & MRA (Magnetic Resonance Angiography)[3]
   - X Requires radiation, and large existing infrastructure[3]
3. **Robotic Ultrasound**
   - ✓ Radiation free
   - X High variability[12]
   - X Requires existing scan data to align[12]

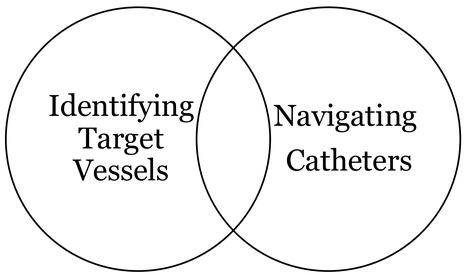


Robotic Guided Ultrasound: Showing actual set up, resultant ultrasound image, and reconstructed 3D view[12]

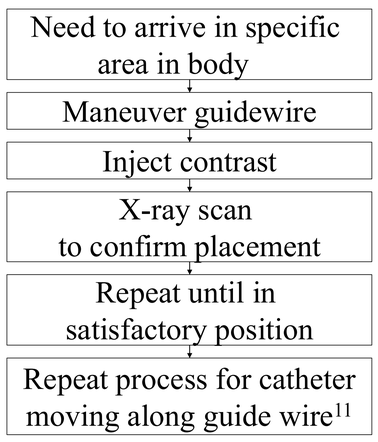
PROBLEM STATEMENT

Interventional radiologists have difficulty guiding catheters through the complex geometry of the patient's vascular system often requiring longer search time/more scans to navigate the catheter.

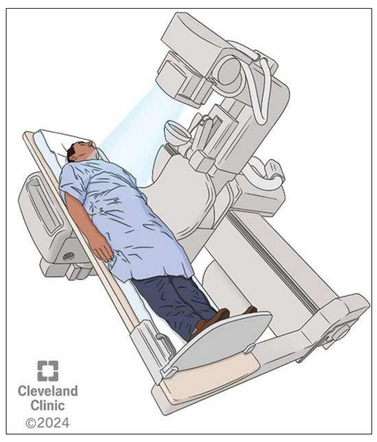
2-Fold Problem:



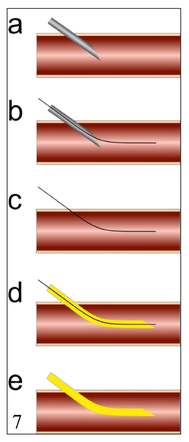
Current Workflow:



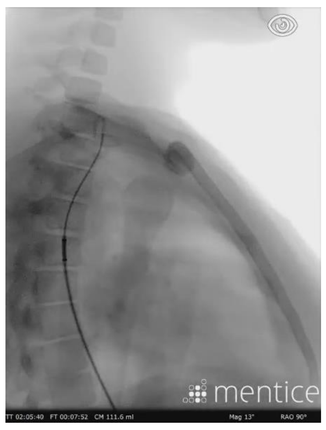
Process Flow



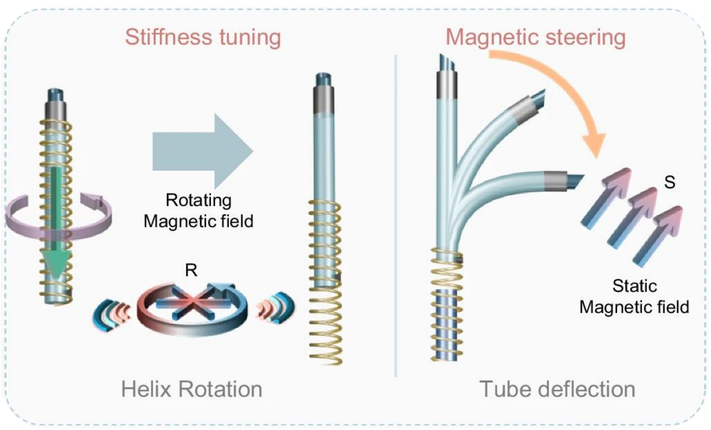
X-ray machine



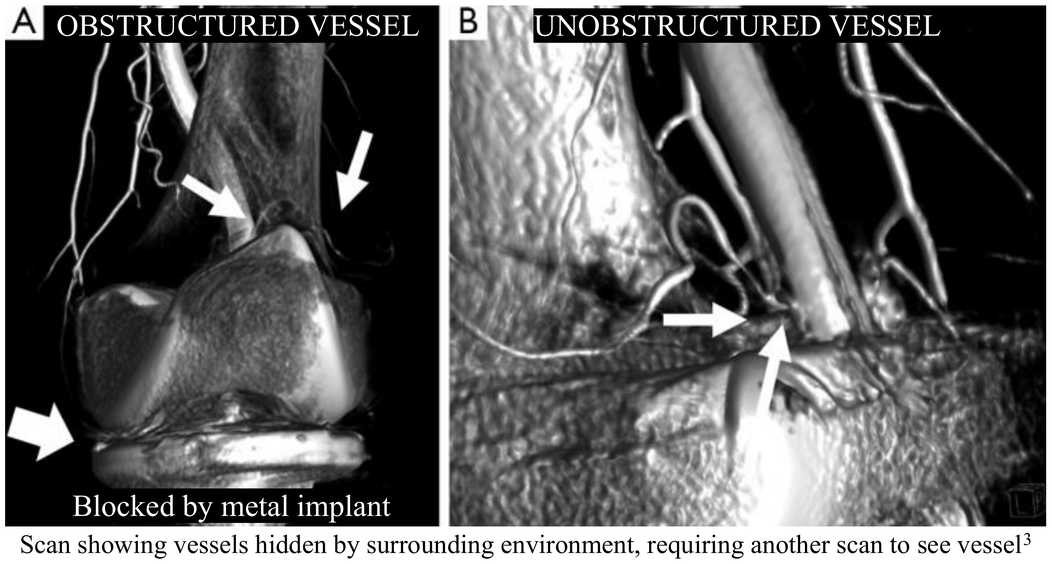
Seldinger Technique



Example X-ray scan with catheter outlined[4]



Working principle of magnetic guidance, allowing active steering and stiffness tuning for different mechanical needs[9]



Scan showing vessels hidden by surrounding environment, requiring another scan to see vessel[3]

Design Constraints

- Compatible with different catheter attachments
- Ingress protection (IPX8)
- Sterilizable
- Legacy tech compatible
- Reimbursable with insurance
- ≤ Training time than current standards

Navigation Success Criteria

- Control all degrees of freedom of catheter
- Biocompatible
- Maintain all functionality of current catheters (delivering medication)
- Easily enter tortuous vessels

Current Navigation Solutions

1. **Seldinger Technique**
   - ✓ Standard practice
   - X Can fail during high curvature turns[8]
   - X Lack of control in all directions[8]
2. **Magnetic Guidance**
   - ✓ Granular control[9]
   - X Requires large infrastructure[9]
3. **Robot Assisted**
   - ✓ Most like current technology
   - X Developing technology[13]

References

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